Homework on String Method equals(), compareTo(), startsWith() endsWith(), and substring()

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A. Multiple Choice Questions (20)

1. Which method is used to compare the *contents* of two strings in Java?
   a) ==
   b) equals()
   c) compare()
   d) match()
2. The == operator in Java compares strings based on:
   a) Content
   b) Length
   c) Reference
   d) Lexicographic order
3. Which method performs case-insensitive string comparison?
   a) equals()
   b) compareTo()
   c) equalsIgnoreCase()
   d) matchIgnoreCase()
4. What does compareTo() return if both strings are equal?
   a) -1
   b) 1
   c) true
   d) 0
5. If s1.compareTo(s2) returns a positive value, then:
   a) s1 < s2
   b) s1 == s2
   c) s1 > s2

d) Compilation error

6. Which method checks whether a string begins with a specific prefix?
   a) begins()
   b) prefix()
   c) startsWith()
   d) endsWith()
7. Which method checks the suffix of a string?
   a) endsWith()
   b) suffix()
   c) stopWith()
   d) finish()
8. Length of string "aaabbbccc" is:
   a) 8
   b) 9
   c) 10
   d) 11
9. substring(startIndex) extracts substring from:
   a) startIndex to end inclusive
   b) startIndex to end exclusive
   c) startIndex inclusive to end
   d) end to start
10. In substring(), start index is:
   a) Exclusive
   b) Inclusive
   c) Optional
   d) Ignored
11. End index in substring(start, end) is:
   a) Inclusive
   b) Exclusive
   c) Optional
   d) Inclusive only if valid
12. Which exception is thrown for invalid

substring index?
a) NullPointerException
b) ArithmeticException
c) StringIndexOutOfBoundsException
d) IndexError

13. s.substring(20) when string length is 20 returns:
a) Error
b) null
c) Empty string
d) Space

14. equals() method belongs to:
a) Object class
b) String class
c) Scanner class
d) Math class

15. Which method is useful for sorting strings?
a) equals()
b) startsWith()
c) compareTo()
d) endsWith()

16. Lexicographic comparison is based on:
a) Length
b) Unicode values
c) Case only
d) Reference

17. s1.equals(s2) returns:
a) int
b) String
c) boolean
d) char

18. Which comparison method is commonly used in authentication?
a) ==

b) equals()
c) compareTo()
d) startsWith()

19. substring(3, 8) extracts how many characters?
a) 3
b) 4
c) 5
d) 6

20. Which of the following is NOT a string comparison method?
a) equals()
b) compareTo()
c) startsWith()
d) substring()

B. Assertion and Reasoning Questions (20)
Directions:
Choose the correct option:
A. Both Assertion and Reason are true and Reason explains Assertion
B. Both Assertion and Reason are true but Reason does not explain Assertion
C. Assertion is true but Reason is false
D. Assertion is false but Reason is true
Assertion: equals() compares the content of strings.
Reason: equals() compares memory references.
Assertion: == compares references in Java strings.
Reason: Strings are objects in Java.
Assertion: compareTo() returns an integer value.
Reason: It performs lexicographic comparison.

Assertion: equalsIgnoreCase() ignores case sensitivity.
Reason: It converts both strings to lowercase internally.
Assertion: substring() may throw an exception.
Reason: Invalid index values cause runtime errors.
Assertion: End index in substring method is inclusive.
Reason: Java string indexing starts from zero.
Assertion: startsWith() checks the prefix of a string.
Reason: It compares characters from the beginning of the string.
Assertion: length() returns the number of characters in a string.
Reason: It returns the size of the string object.
Assertion: substring(20) is valid if the string length is 20.
Reason: Start index can be equal to string length.
Assertion: compareTo() is useful in sorting strings.
Reason: It provides ordering information.
11–20. (Continue similar pattern for practice or examination.)

C. Output Based Questions (20)
Predict the output of the following code snippets:

```
String s1 = "Java";
String s2 = "Java";
System.out.println(s1.equals(s2));
```

```
String s1 = new String("Java");
String s2 = new String("Java");
System.out.println(s1 == s2);
String s = "aaabbbccc";
System.out.println(s.length());
String s = "Computer Application";
System.out.println(s.substring(9));
System.out.println("Ram".equalsIgnoreCase("rAm"
));
System.out.println("abc".compareTo("abc"));
System.out.println("abc".compareTo("abd"));
System.out.println("abd".compareTo("abc"));
System.out.println("aaabbbccc".startsWith("aaa"
));
System.out.println("aaabbbccc".endsWith("bbb"))
;
```

11–20. (More similar output-based questions may be added.)

D. True / False Questions (10)

1. == compares string content.
2. equals() returns a boolean value.
3. compareTo() can return a negative value.
4. Substring end index is inclusive.
5. startsWith() checks suffix.
6. length() returns an integer.
7. equalsIgnoreCase() is case sensitive.
8. substring() may throw an exception.
9. Strings are objects in Java.
10. compareTo() is used for sorting strings.

E. Fill in the Blanks (10)

1. equals() compares string _____.
2. == compares string _____.

3. compareTo() returns an _____ value.
4. equalsIgnoreCase() ignores _____.
5. Substring start index is _____.
6. Substring end index is _____.
7. The exception thrown is _____.
8. startsWith() checks the _____.
9. endsWith() checks the _____.
10. length() returns the number of _____.